

**A Multi-Modal Deep Learning Pipeline Integrating Physics-Informed Neural Networks
and 3D Image Processing for Accurate Rupture Prediction and Data-Driven Surgical
Planning of Cerebral Aneurysms: Year 3**

Abstract

Background: Cerebral aneurysm is an abnormal bulging of the arterial wall that poses significant risk to patient health due to the potential for rupture and brain hemorrhage. Clinical treatment of cerebral aneurysms entails highly invasive endovascular surgery, which in itself carries risk of patient mortality. Therefore, there is a need to accurately quantify the severity of the disease (i.e. risk of rupture) to enable more informed surgical planning and treatment. To achieve this goal, prior work focused on developing CFD-based web and virtual-reality tools to visualize and quantify blood flow patterns in normal and diseased arteries; however, derivation and proper interpretation of flow data to accurately determine aneurysm rupture risk remains challenging.

Methods: Year 3 of this work focused on building an end-to-end diagnosis pipeline to detect and estimate the rupture probability of cerebral aneurysms using three primary deep learning models. The first PointNet++ model detects the presence of aneurysm using an open-source dataset of different vascular geometries. Next, an unsteady physics-informed neural network (PINN) derives flow patterns in the artery using physical constants from the Navier-Stokes partial differential equations and initial conditions. Lastly, a multichannel PointNet++ model interprets the flow patterns towards an informed prediction of aneurysm rupture.

Results: Statistical analysis using Area Under Curve (AUC) revealed strong success with aneurysm detection (AUC = 0.95). Loss functions indicate that the PINN correction models closely replicate the governing laws of the Navier-Stokes equations (physics loss $< 1 \times 10^{-13}$). A series of modeling strategies were tested to predict rupture prediction, yielding maximum AUC = 0.75.

Conclusion: Based on the results of this work, combining physics-informed neural networks and deep learning techniques is a viable strategy to accurately predict aneurysm rupture. Future work will focus on integrating additional parameters such as wall thickness and relative residence time in these deep learning models to maximize accuracy.

**2026 Brevard Regional Science & Engineering Fair
General Research Plan Form**

Name	
School	West Shore Junior/Senior High School
Category	Mathematics and Computational Sciences
Research Teacher	

A. Title

A Multi-Modal Deep Learning Pipeline Integrating Physics-Informed Neural Networks and 3D Image Processing for Accurate Rupture Prediction and Data-Driven Surgical Planning of Cerebral Aneurysms: Year 3

B. Question or Problem Being Addressed

Cerebral aneurysm rupture diagnosis is highly dependent on clinical analysis of qualitative scans. Computational fluid dynamics offers a solution to this by quantifying blood flow data, but is limited by extensive computational power requirements. Therefore, there is a need for a neural network driven approach to quantify blood flow data in an effective manner, aiding diagnosis of rupture probability.

C. Hypothesis/Engineering Goals

1. Develop a 3D CNN model to diagnose cerebral aneurysms from 3D geometry data.
2. Integrate physics-informed neural networks to improve diagnostic performance of neural network models.
3. Visualize 3D solution data alongside prediction from neural network models to aid clinical diagnosis.

D. Variables (Independent and Dependent)

N/A

E. Rationale - Brief synopsis of the background that supports your research problem and explain why this research is important scientifically, and if applicable, explain any societal impact of your research.

Cerebral aneurysms (CA) are a major issue in the health of the people between the ages of 30 and 60, and affect nearly 5% of the US population. Although most cases lack severity and are asymptomatic, aneurysm rupture can quickly become life-threatening through hemorrhage and stroke. The current goal standard for aneurysm diagnosis, an angiography, is insufficient for accurate diagnosis of aneurysm severity, creating a

need for a quantitative diagnostic methodology.

In order to combat this problem, neural network models can be used to analyze 3D aneurysm geometries and determine rupture probability. This study proposes a physics-informed neural network to improve the diagnostic ability of neural network models. These models use fluid flow conditions to calculate the physics variables (e.g. wall shear stress) to quantitatively analyze the 3D geometry of a cerebral aneurysm and combine the data with the physical characteristics (e.g., size, shape) to create a holistic rupture diagnostic.

F. Materials List - List of all items used in research. Make sure to include concentrations of all chemicals, source and amount of all living organisms, and all equipment used. The list should be bulleted or numbered.

Item	Source
Laptop Computer	Researcher
AI.Panther Supercomputer	Florida Institute of Technology

G. Procedures

Safety Plan

1. In order to mitigate the risks associated with long-term exposure to blue light, the 20-20-20 rule will be implemented. The rule indicates that a 20 second break will be taken every 20 minutes spent on the computer, while looking at something 20 feet away.
2. No audio will be utilized in the application

Experimentation Steps (Safety needs to be included throughout the experimentation steps)

1. In order to mitigate the risks associated with long-term exposure to blue light, the 20-20-20 rule will be implemented. The rule indicates that a 20 second break will be taken every 20 minutes spent on the computer, while looking at something 20 feet away.
2. Obtain deidentified patient data (<https://zenodo.org/records/6678442>)
 - a. Data is found under files.
3. Develop 3D CNN and Swin-transformer model to serve as a baseline architecture.
 - a. Existing Swin-transformer architecture can be found here: [https://github.com/JZK00/3D Swin Transformer Classification](https://github.com/JZK00/3D_Swin_Transformer_Classification)
4. Integrate physics-informed neural networks to improve model accuracy.
5. Run all model designs on the AI.Panther server hosted at the Florida Institute of Technology.
6. Develop a final application that can display 3D CFD data and model predictions in a single user interface.
7. Experimentation will be done at home under the supervision of the designated supervisor (see Form 3). Florida Institute of Technology will be used as a remote work site with mentorship from Dr. Xianqi Li (will not directly supervise, but will provide mentorship).

**Disposal (What are the specific procedures? Even if you aren't responsible for them).
Disposal of chemicals and other materials will need to also be listed.**

1. N/A

H. Data Analysis: Describe the procedures you will use to analyze the data that answers research questions, hypotheses, or engineering goals. Steps should be numbered.

1. Compare the accuracy of the two models (see experimentation steps) using various statistical measures (e.g. root mean squared error)
2. The engineering goals will be met if the final model has a 90% prediction accuracy.

I. Bibliography - Specific citations needed depending on your project--human subjects, vertebrate animal care, microbiology protocol, etc. A total of five full citations are required for approval. Must be formatted in MLA/APA. See rule book for specific resources.

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<https://doi.org/10.1117/12.2653871>

SDS Citations: Cite all chemicals used here using the specific chemical's SDS form (not a general link to all SDS's) . Must be formatted in MLA/APA.They should also be found on Form 3.

1. N/A

Introduction

Cerebral aneurysms, a disorder that describes the dilation of blood vessels in the brain, are a leading cause of death in the United States (Mayo Foundation for Medical Education and Research, 2023). Diagnosis of cerebral aneurysms is complex, and treatment often involves highly invasive surgery associated with significant risk to the patient (Geers et al., 2011). Due to the large degree of uncertainty surrounding the mortality, rupture risk, and severity of cerebral aneurysms, understanding the prognosis (end result) of cerebral aneurysms is vital in order to minimize risk and avoid unnecessary treatment.

Estimating the rupture risk of cerebral aneurysms relies on angiographic injection, a process which utilizes a contrast medium coupled with an angiogram to view the hemodynamics (blood flow) in an artery, and qualitative analysis by clinicians (Moore et al., 2017). Novel technologies such as Computational Fluid Dynamics (CFD) offer quantitative diagnostic solutions; however, these methodologies require clinicians to acquire significant technical expertise and are inefficient in time and cost (Frösen, 2016; White et al., 2023).

Deep learning (DL) is a rising technology capable of estimating rupture risk of cerebral aneurysms by analyzing wall structure and aneurysm size. Diagnostic accuracy can be enhanced by combining the physical structure of cerebral aneurysms with hemodynamic parameters that describe the blood flow within the artery. Parameters such as Velocity Vector Magnitude (VVM), Wall Shear Stress (WSS), Oscillatory Shear Index (OSI), and Von Mises Stress (VMS) can aid DL frameworks such as PointNeXt in accurately estimating the rupture risk of cerebral aneurysms. These parameters can be obtained by training physics-informed neural networks (PINNs) that incorporate physical constants of the Navier-Stokes equations to estimate blood flow within an artery. Therefore, developing an end-to-end workflow that combines

hemodynamic parameters derived from PINNs as input for DL frameworks with features based on aneurysm geometry and patient history can form an accurate holistic benchmark for aneurysm rupture risk.

Literature Review

Basic Information

A cerebral aneurysm (CA) describes the bulging or ballooning of an artery in the brain. CAs are known to impact those around the age of 50, with a larger connection to males over females. An estimated 6.7 million people in the US have an unruptured cerebral aneurysm, but most are undiagnosed due to the lack of symptoms and the complex diagnosis methods of the disease (Brain Aneurysm Foundation, 2023). Only 1-2% of cerebral aneurysms rupture each year due to damage of endothelial cells on the arterial wall, but ruptured aneurysms can cause hemorrhage and patient death if improperly treated (Gimbrone, 1999). Treatment of cerebral aneurysms involves an intracranial angioplasty, subjecting patients to the risks of brain surgery and creating a need for accurate diagnosis (Geers et al., 2011).

Existing Diagnostic Methodologies

The primary method for diagnosis of cerebral aneurysms is angiographic injection. The method entails the injection of a contrast medium into the bloodstream and the subsequent monitoring of how the dye travels through vasculature with an X-Ray or magnetic resonance image (Abdollahi et al., 2021). Analysis of this 2D/3D angiographic imaging yields a black-and-white animation of the contrast medium. The image/video can be used to identify aneurysms, stenosis, and other vascular conditions; however, concluding the rupture risk of the condition is difficult with strictly qualitative information of blood flow (White et al., 2023).

Because of this, the bulk of estimating the rupture probability of cerebral aneurysms lies with the clinician, resulting in an error-prone process.

An example of current diagnostic methodologies is the PHASES score. The PHASES score combines the parameters of population (demographic), hypertension, aneurysm size/location, and earlier subarachnoid hemorrhage to predict aneurysm rupture probability. However, patient studies over a long period of time have shown the ineffectiveness of this method, exemplifying the difficulty of estimating rupture risk.

Computational Fluid Dynamics

Computational Fluid Dynamics is a physics-based approach to increasing the quantitative data available to clinicians (Lee, 2011). By employing the Navier-Stokes Equations and complex equation-solvers, quantitative hemodynamic (blood flow) data can be derived from a 3D rotational angiography (Jeong & Rhee, 2012; Yagi et al., 2020). The data can be in the form of Velocity Vector Magnitude (VVM), Wall Shear Stress (WSS), Oscillatory Shear Index (OSI), and Von Mises Stress (VMS) (Han, 2021).

VVM describes the velocity of specific points in the artery. Narrowed regions of the artery would have higher VVM, placing more stress on the wall, while expanded regions would have lower VVM, symbolizing inadequate blood flow (White et al. 2023). WSS represents the shear stress placed on the wall, serving as a direct measure for rupture risk. A higher WSS in narrowed regions of the artery (stenosis) signifies a higher rupture risk. Imbalance in WSS has been shown to cause damage to the endothelial cells, a direct correlation to aneurysm rupture. OSI explains the reversal of blood flow in the artery, caused by abnormal arterial shapes. OSI applies most for aneurysm diagnosis as the expanded arterial wall causes the reversal of blood flow, placing stress on the arterial wall and providing inadequate blood flow. VMS represents the

actual stress placed on the wall of an artery given varying levels of stress over a period of time. This is clinically relevant because the pulsatile nature of blood flow (systole and diastole) places varying amounts of stress on the arterial wall.

CFD data can be obtained through simulation with commercial software such as Ansys Fluent or open-source alternatives such as OpenFOAM and FreeFEM++. Prior to running such simulations, data from aneurysm datasets must be converted to watertight volumetric meshes via a rigorous meshing process in softwares such as Gmsh. The solving of Navier-Stokes (and other such partial differential equations) can be achieved through three main methods: finite difference, a technique that approximates the derivatives of a partial differential equation using numerical differences; finite volume (OpenFOAM & Ansys Fluent), a technique that ensures conservation of mass and momentum, while dividing the mesh into smaller control volumes; and finite element analysis (FreeFEM++), a technique that simplifies the terms of partial differential equations into finite elements.

Physics-Informed Neural Networks

Through physics-informed neural networks (PINNs), efficient solvers for the Navier-Stokes partial differential equations can be developed. These solvers allow for rapid dissemination of hemodynamic parameters to be deployed in DL frameworks. Fusing hemodynamic parameters with structural and patient history features can create a more informed diagnostic measure.

Forming a PINN requires an existing sample set of solved CFD cases. To achieve this, open-source software such as Gmsh, OpenFOAM, and FreeFEM++ can be used to simulate blood flow with a pulsatile input velocity, given a clearly defined inlet and outlet(s). The largest opening (by cross-sectional area) can be assumed to be the inlet, as done in prior works.

Deep Learning

PointNeXt is a deep learning framework that extracts features about the structural integrity of a cerebral aneurysm by analyzing point clouds. This methodology is different from methods such as convolutional neural networks that use voxels to represent a cerebral aneurysm volume. PointNeXt can be configured to accept multi-channel inputs, allowing hemodynamic parameters to be used as features.

Data from PointNeXt can be combined with non-imaging parameters such as age, biological sex, and prior history through XGBoost, forming a fusion model that predicts the rupture probability of cerebral aneurysms.

Problem

Cerebral aneurysms (CA) are a major issue in the health of the people between the ages of 30 and 60, and affect nearly 5% of the US population. Although most cases lack severity and are asymptomatic, aneurysm rupture can quickly become life-threatening through hemorrhage and stroke. The current gold standard for aneurysm diagnosis, an angiography, is insufficient for understanding aneurysm severity, creating a need for a quantitative diagnostic methodology.

In order to combat this problem, neural network models can be used to analyze 3D aneurysm geometries and determine rupture probability. This study proposes a complete deep learning pipeline to accurately estimate CA rupture probability through physics-informed neural networks (PINNs) and analysis of 3D aneurysm geometries.

Engineering Goals

1. Develop an aneurysm detection model using PointNeXt.
2. Build a physics-informed neural network to decrease CFD simulation time by $> 90\%$.
3. Use flow parameters in neural network models to accurately predict aneurysm rupture.
4. Visualize fluid flow data through web and VR applications.

Methods

DeepONet/Physics-Informed Neural Networks (PINNs)

To generate the flow fields, velocity and pressure data for the steady state flow was obtained from Anuemo. Data preprocessing correctly altered the VTP data from the AneuX dataset in order to use the DeepONet inference scripts.

The velocity/pressure fields from DeepONet were used as a reference in a PINN-correction script which directly reduced the physics-error by incorporating Navier-Stokes terms in the loss function, a process known as regularization. The PINN-correction outputted unsteady data (time-dependent), allowing for more accurate representation of pressure and velocity as well calculation of advanced hemodynamic parameters (wall shear stress, oscillatory shear index, and von Mises stress).

Modelling Strategy

The diagnostic methodology relied on PointNeXt, a sparse point cloud processing algorithm. The multichannel feature of PointNeXt can be used to include flow features in model processing. Therefore, three different models were trained: 1) a geometry-based model that considers the x, y, z spatial coordinates; 2) a basic flow-based model that considers spatial coordinates, pressure, and velocity vectors; and 3) an advanced flow-based model that considers spatial coordinates, pressure, velocity vectors, wall shear stress vectors, oscillatory shear index, and von Mises stress.

The three models are run on the AneuX dataset with balanced classes, resulting in a total of ~500 cases. A training-validation split of 80-20 was used, resulting in ~400 training cases and ~100 validation cases. All models were run on the AI.Panther server for 200 epochs, an Adam learning optimizer, and a learning rate of 1e-3.

To analyze data, metrics such as Receiver Operating Characteristic (ROC) curve, AUC, accuracy, loss, and confusion matrix were measured and analyzed.

Summary of Model Architectures	
Model	Architecture
Detection	PointNeXt with 4 layers
Physics-Informed Neural Network	Multilayer perceptron with 4 layers
Rupture Prediction	Multichannel PointNeXt with multilayer perceptron - geometry/flow each have 3 set abstraction layers that contain 3 Conv2d layers (1 for flow has 2) - clinical encoder has 2 layers - fusion head has 3 linear layers - global feature pooling has 1 layer - 23 layers in total

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<https://doi.org/10.1016/j.jcp.2018.10.045>

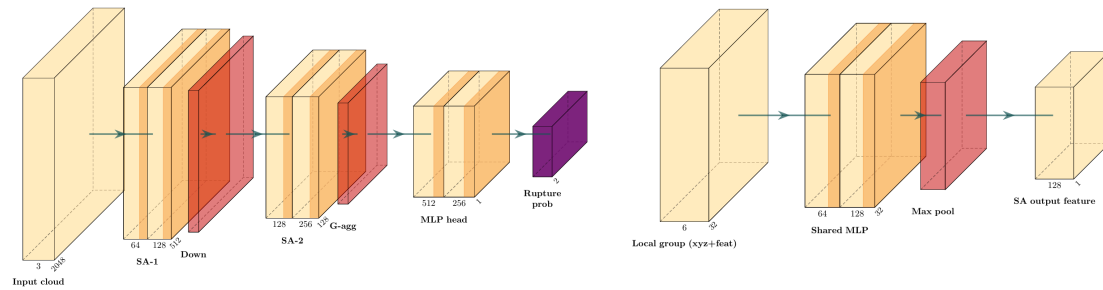
Rehman, M. A. R., Ekici, Ö., Farooq, M. A., Butt, K., Ajao-Olarinoye, M., Wang, Z., & Liu, H. (2025). Fluid–structure interaction analysis of pulsatile flow in arterial aneurysms with physics-informed neural networks and computational fluid dynamics. *Physics of Fluids*, 37(3), Article 031913. <https://doi.org/10.1063/5.0259296>

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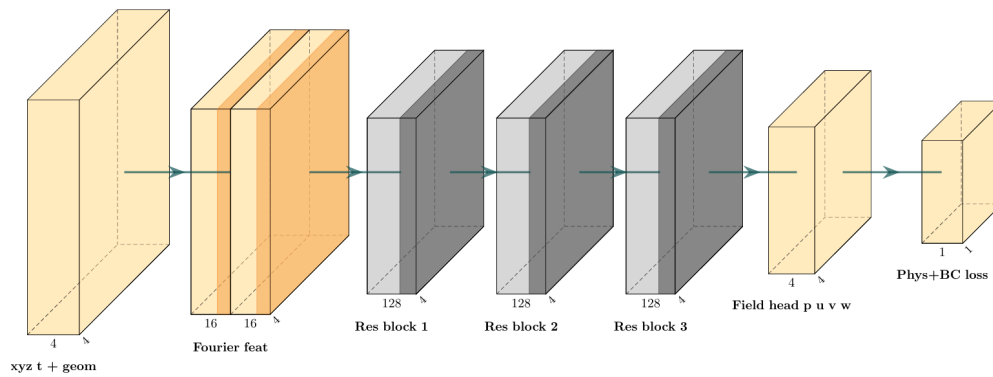
Shou, Y., Chen, Z., Feng, P., Wei, Y., Qi, B., Dong, R., Yu, H., & Li, H. (2024). Integrating pointnet-based model and machine learning algorithms for classification of rupture status of IAs. *Bioengineering*, 11(7), 660. <https://doi.org/10.3390/bioengineering11070660>

Model Architectures

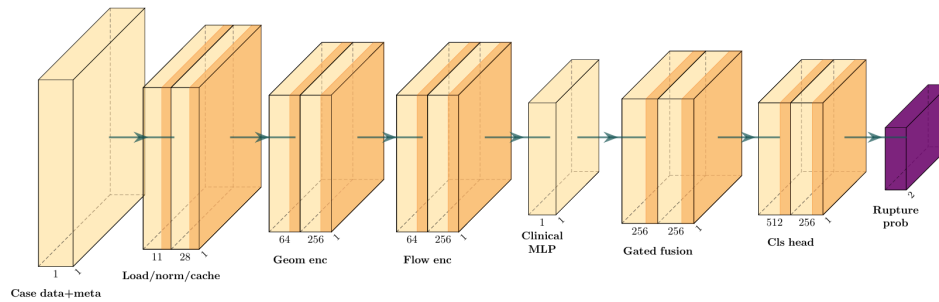
Detection Model



PINN



Rupture Prediction

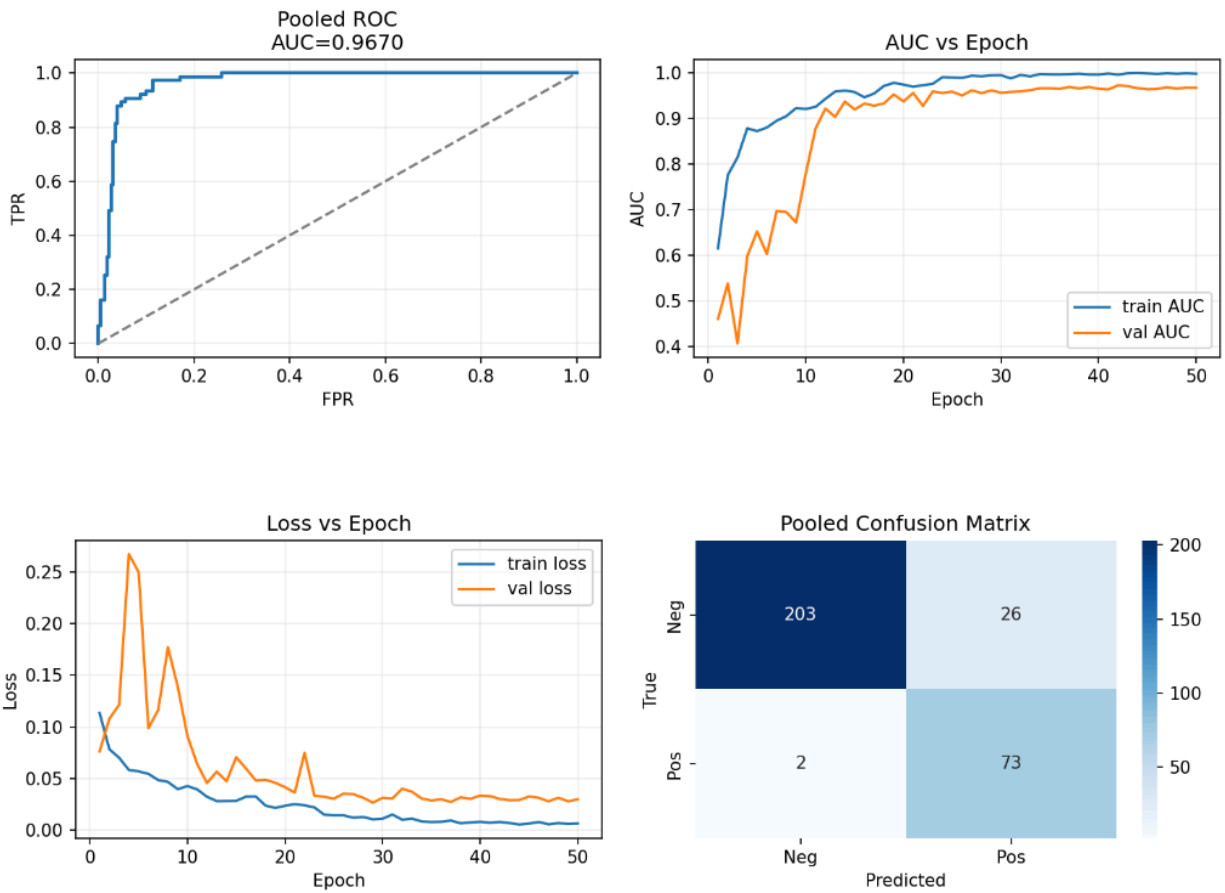


Graphs created using Python and LaTeX with source code from PlotNeuralNet:

<https://github.com/kgruiz/PlotNeuralNet>

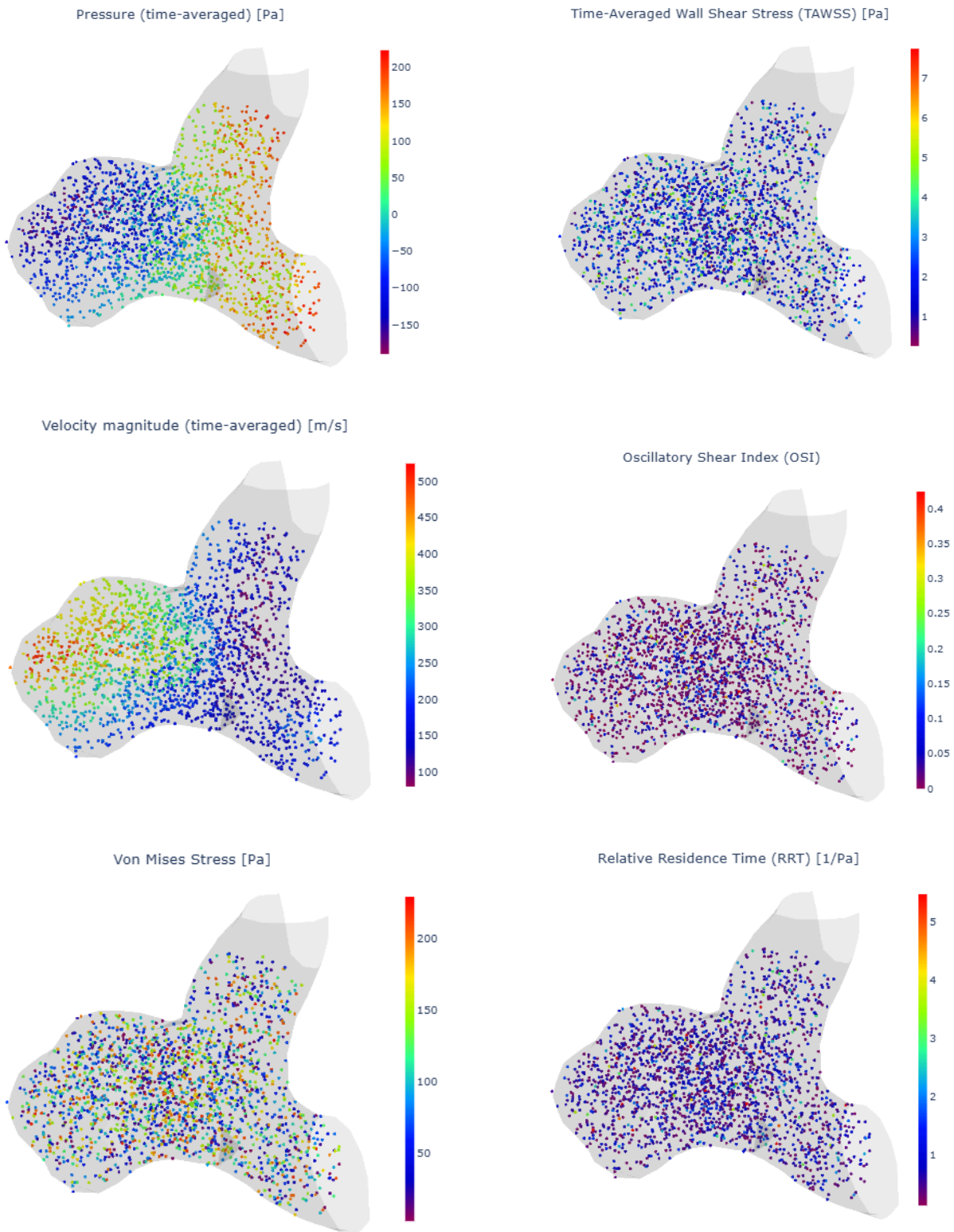
Data

Aneurysm Detection

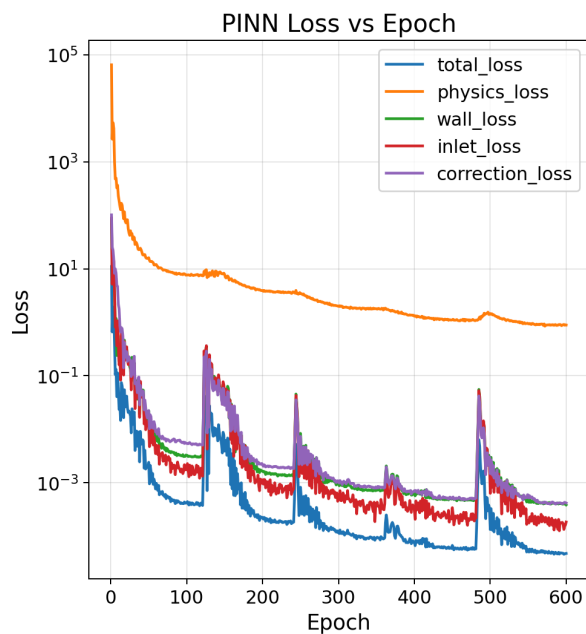


Graphs produced by researcher using Python and matplotlib (2026).

Physics Informed Neural Networks



Graphs generated by researcher using plotly (2026).



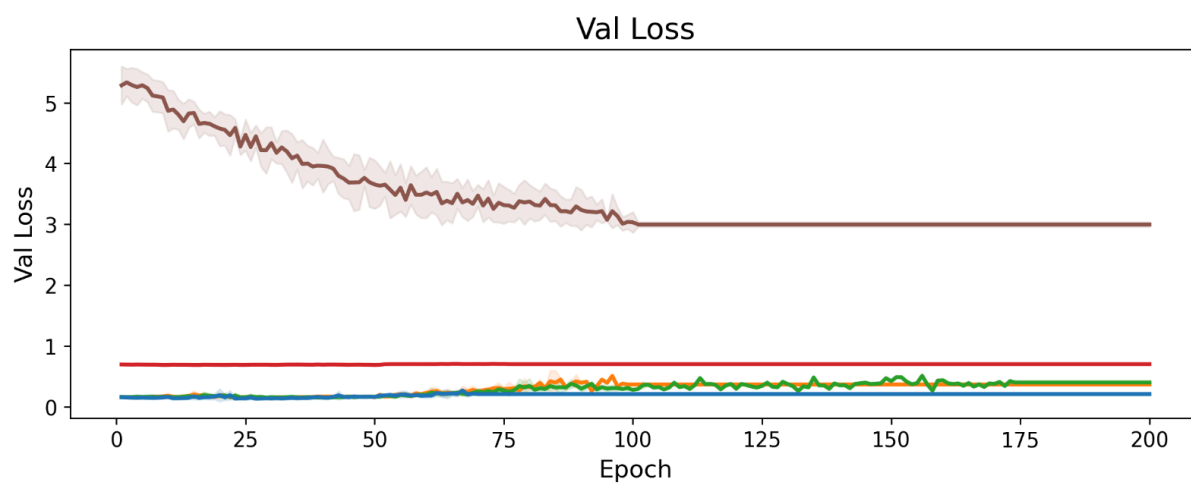
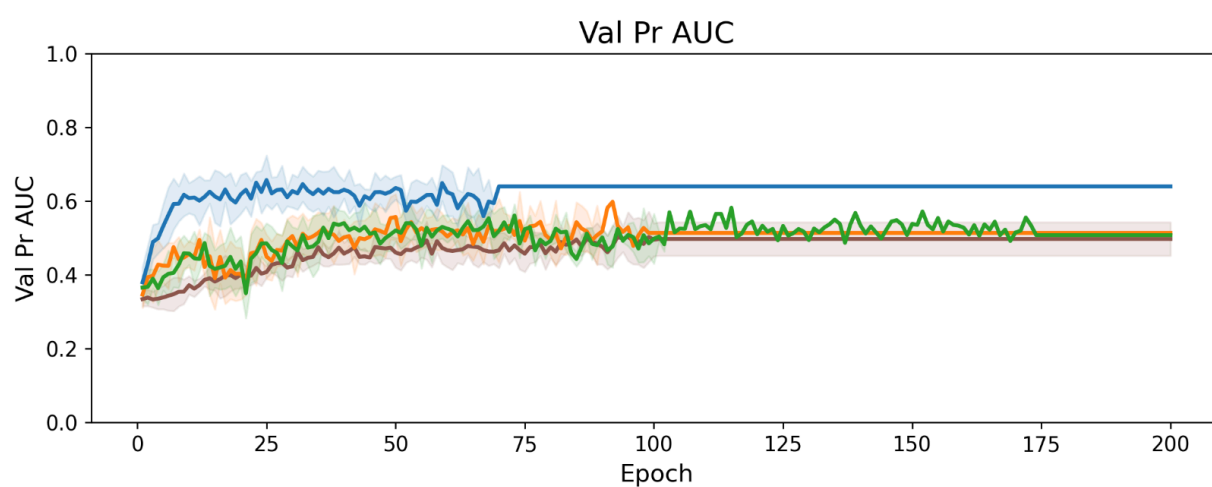
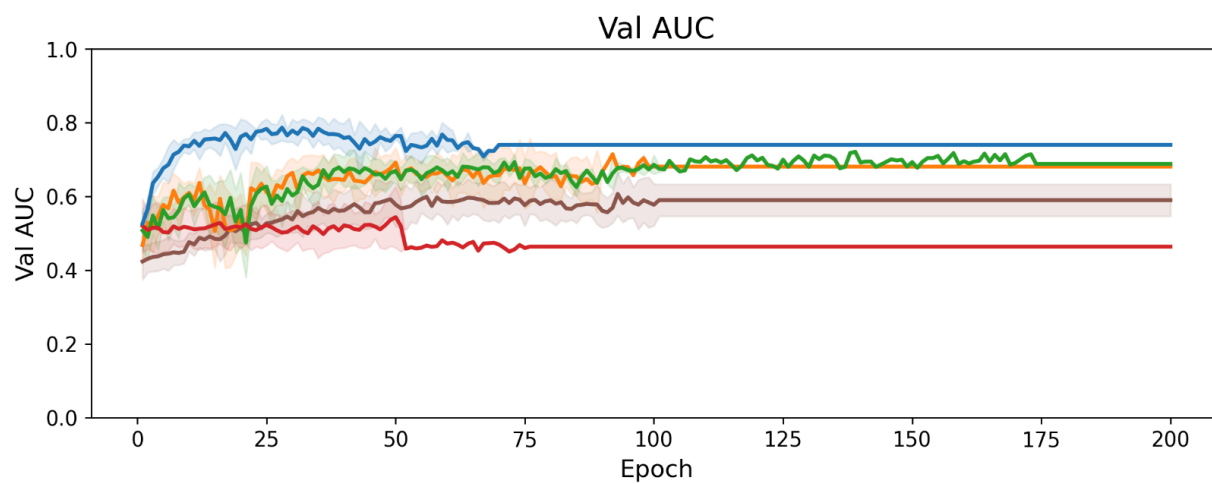
Graph generated by researcher using Python and matplotlib (2026).

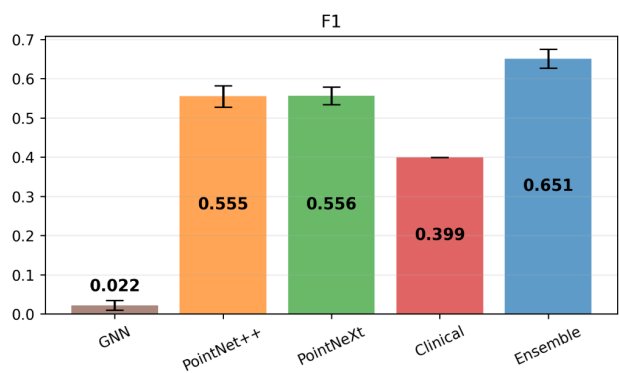
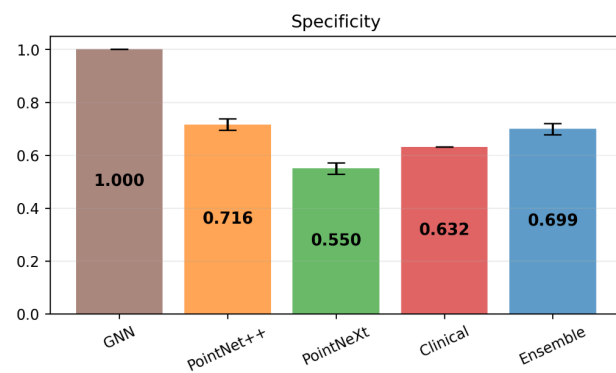
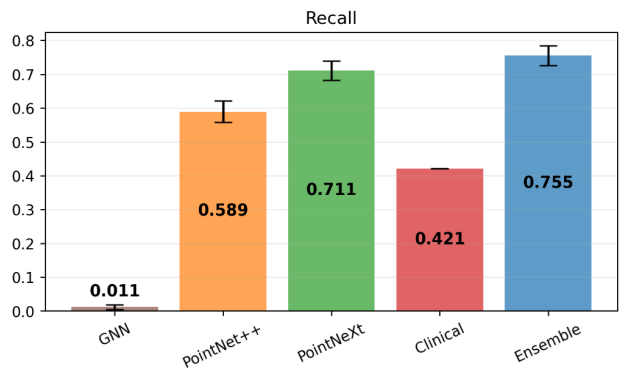
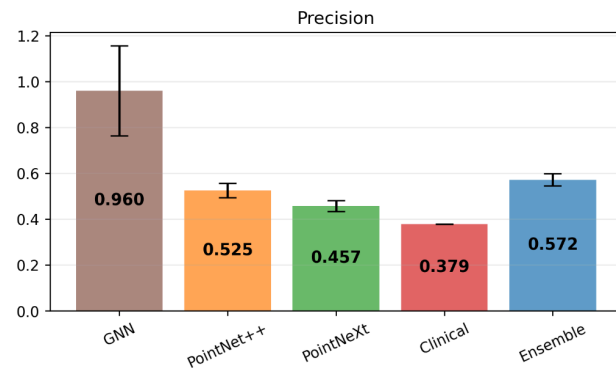
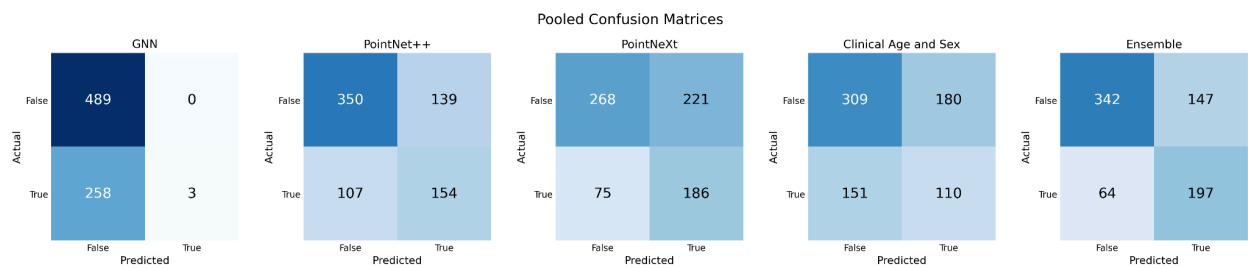
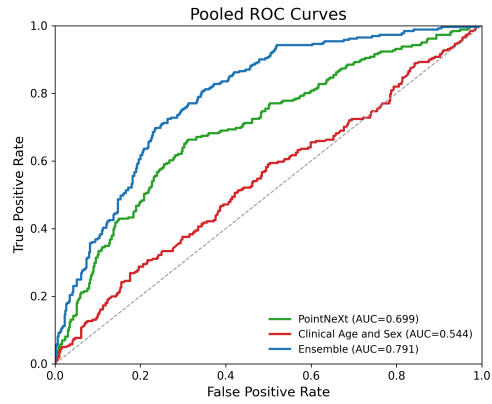
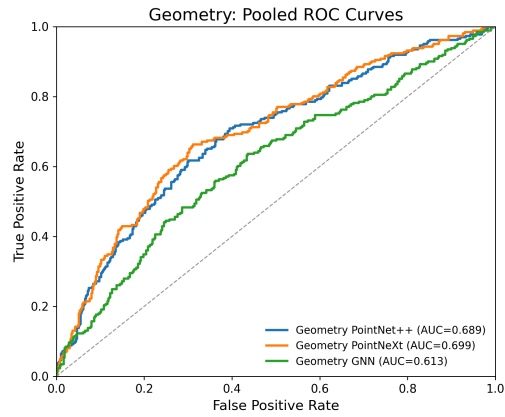
Prediction Modeling

Architecture	Val AUC
GNN	0.613
PointNeXt	0.689
PointNeXt	0.699

Architecture	Train Acc	Train AUC	Val Acc	Val AUC
Clinical	53.1%	0.532	55.7%	0.561
Geometry	65.2%	0.718	60.7%	0.699
Ensemble	73.3%	0.797	71.7%	0.791

Tables created by researcher using Google Slides (2026).





Graphs created by researcher using Python and matplotlib (2026).

Results

The final deliverable of the study was a pipeline that can detect, simulate, and diagnose cerebral aneurysms. The first model in the pipeline uses PointNeXt and data from the Vascular Model Repository to detect whether a specific geometry is an aneurysm. This model had a final AUC of 0.967 and accuracy of 92.1%, indicating strong success. Upon detection of an aneurysm, the pipeline simulates the flow patterns in the artery using physics-informed neural networks (PINN) based on the Navier-Stokes equations, yielding an accurate flow field with an average training time of less than six minutes per case. From the PINN correction, velocity, wall shear stress, and pressure were derived over time, allowing for calculation of oscillatory shear index, von Mises stress, and relative residence time.

These six hemodynamic parameters were fused with x, y, z positional data as input to a multichannel PointNeXt model in order to estimate rupture probability. As a baseline, a PointNeXt model was trained without any flow parameters. Representing diagnosis without deep learning techniques, the clinical model interpreted age, biological sex, and location of aneurysm to determine rupture risk (AUC = 0.561). To evaluate the performance of the three modeling strategies, k-fold cross validation was implemented with 80% of the dataset used for training and 20% used for validation on each of the five folds. Average AUC across the five folders revealed superior performance for the PointNeXt model with flow data (AUC = 0.791) when compared to the baseline (AUC = 0.699) and clinical (AUC = 0.561) models.

Conclusions

All four of the stated engineering goals were met. Based on the AUC of 0.967, aneurysm detection was successful. Since the features extracted by the deep learning model (cross-sectional area, irregular shape, etc) are known by clinicians, the initial model serves as an automation for existing clinician labor. As an intermediate step, PINN correction algorithms significantly improved the speed of fluid physics calculations as traditional computational fluid dynamics techniques require hours to days of simulation time, resulting in better scalability for large datasets. Lastly, the rupture prediction model differentiates from existing clinicians as it provides better clarity in understanding of rupture probability. Therefore, the AUC of 0.79 offers a significant improvement over current standards (e.g. the PHASES score which has been shown to have low accuracy). The fluid physics and rupture prediction data was displayed to the end user in a user-friendly web application, with the option to visualize datasets in virtual-reality. Overall, the PINN-based pipeline developed in this project offers a quantitative rupture prediction metric for clinicians towards more informed surgical interventions.

Limitations/Future Work

1. Time-Dependency: AneuX only provides ground truth for whether the aneurysm ruptured, not how the disease progressed over time. Further modeling could be used to forecast transient aneurysm growth and rupture.
2. Additional Hemodynamic Parameters: The calculation of additional parameters such as wall thickness, mean transit time, and relative residence time can facilitate further analysis of blood flow patterns.
3. Surgical Risk: Integrating risk factors of endovascular surgery with aneurysm rupture probability can provide additional insight into surgical planning.